Chapter 1: The Real and Complex Number Systems

Section: Fields

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

The Field Axioms

Remarks and Notation

Consequences of the Field Axioms

Ordered Fields

Motivation

Why This Section Matters

We need a precise algebraic framework for the number systems used in analysis

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The rationals $\mathbb{Q}$, the reals $\mathbb{R}$, and the complex numbers $\mathbb{C}$ are all *fields*

Adding *order* to a field gives an ordered field — the setting for inequalities

The Field Axioms

Definition 1.12: Field

Definition 1.12

A field is a set F with two operations, *addition* and *multiplication*, satisfying three groups of axioms:

(A) Axioms for addition

(M) Axioms for multiplication

(D) The distributive law

Definition 1.12: Axioms for Addition

(A) Axioms for Addition

(A1) If $x, y \in F$, then $x + y \in F$ *(closure)*

(A2) $x + y = y + x$ *(commutativity)*

Definition 1.12: Axioms for Addition

(A) Axioms for Addition

(A1) If $x, y \in F$, then $x + y \in F$ *(closure)*

(A2) $x + y = y + x$ *(commutativity)*

(A3) $(x + y) + z = x + (y + z)$ *(associativity)*

(A4) $\exists\, 0 \in F$ such that $0 + x = x$ for all x *(zero element)*

Definition 1.12: Axioms for Addition

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(A1)	If $x, y \in F$, then $x + y \in F$	*(closure)*
(A2)	$x + y = y + x$	*(commutativity)*
(A3)	$(x + y) + z = x + (y + z)$	*(associativity)*
(A4)	$\exists\ 0 \in F$ such that $0 + x = x$ for all x	*(zero element)*
(A5)	For every $x \in F$, $\exists\ -x \in F$ with $x + (-x) = 0$	*(additive inverse)*

Definition 1.12: Axioms for Multiplication

(M) Axioms for Multiplication

(M1) If $x, y \in F$, then $xy \in F$ *(closure)*

(M2) $xy = yx$ *(commutativity)*

Definition 1.12: Axioms for Multiplication

(M) Axioms for Multiplication

(M1)	If $x, y \in F$, then $xy \in F$	*(closure)*
(M2)	$xy = yx$	*(commutativity)*
(M3)	$(xy)z = x(yz)$	*(associativity)*
(M4)	$\exists\, 1 \in F$, $1 \neq 0$, such that $1x = x$ for all x	*(unity)*

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(M3)	$(xy)z = x(yz)$	*(associativity)*
(M4)	$\exists\ 1 \in F$, $1 \neq 0$, such that $1x = x$ for all x	*(unity)*
(M5)	If $x \neq 0$, $\exists\ 1/x \in F$ with $x \cdot (1/x) = 1$	*(mult. inverse)*

Definition 1.12: The Distributive Law

(D) The Distributive Law

For all $x, y, z \in F$:

$$x(y + z) = xy + xz$$

The distributive law is the bridge between addition and multiplication.

Field Axiom Structure

(A) Addition
A1–A5
Abelian group

(M) Multiplication
M1–M5
(nonzero elements)
Abelian group

(D) Distributive Law
$x(y+z) = xy + xz$

Remarks and Notation

Remarks 1.13: Shorthand Notation

Remark 1.13(a): Notation

In any field, we write:

$$x - y \text{ for } x + (-y), \qquad \frac{x}{y} \text{ for } x \cdot \frac{1}{y}$$

$$x^2 \text{ for } xx, \qquad 2x \text{ for } x + x, \qquad \text{etc.}$$

Remarks 1.13: $\mathbb{Q}$ Is a Field

Remark 1.13(b)

The rational numbers $\mathbb{Q}$, with the usual addition and multiplication, form a field.

All ten axioms (A1)–(A5), (M1)–(M5), and (D) hold in $\mathbb{Q}$.

Remarks 1.13: Why Study Fields?

Remark 1.13(c)

Familiar properties of $\mathbb{Q}$ are *consequences* of the field axioms.

Key insight: Proving results at the field level means we get them for free in $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$.

Consequences of the Field Axioms

Proposition 1.14: Cancellation Laws for Addition

Proposition 1.14

The axioms for addition imply:

(a) If $x + y = x + z$ then $y = z$ *(cancellation law)*

(b) If $x + y = x$ then $y = 0$ *(uniqueness of 0)*

Proposition 1.14: Cancellation Laws for Addition

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(b) If $x + y = x$ then $y = 0$ *(uniqueness of 0)*

(c) If $x + y = 0$ then $y = -x$ *(uniqueness of $-x$)*

(d) $-(-x) = x$ *(double negation)*

Proof of Proposition 1.14 — Strategy

Goal: Prove (a), then derive (b), (c), (d) from (a).

Key idea for (a): Add $-x$ to both sides and use the axioms to simplify.

$$\begin{tikzcd}
(a) \arrow[r, "z = 0"] \arrow[rr, bend left, "z = -x"] & (b) & (c) \arrow[r, "-x \text{ for } x"] & (d)
\end{tikzcd}$$

Proof of Proposition 1.14(a)

Proof of (a): If $x + y = x + z$ then $y = z$

Assume $x + y = x + z$. Then:

$$
\begin{aligned}
y &= 0 + y && \text{by (A4)} \\
&= (-x + x) + y && \text{by (A5)} \\
&= -x + (x + y) && \text{by (A3)} \\
&= -x + (x + z) && \text{by hypothesis} \\
&= (-x + x) + z && \text{by (A3)} \\
&= 0 + z = z && \text{by (A5), (A4)}
\end{aligned}
$$

Proof of Proposition 1.14(b)–(d)

Recall 1.14

(a) $x + y = x + z \Rightarrow y = z$

(b) $x + y = x \Rightarrow y = 0$

(c) $x + y = 0 \Rightarrow y = -x$

(d) $-(-x) = x$

Deriving (b), (c), (d) from (a)

(b): $x + y = x = x + 0$, so $y = 0$ by (a) with $z = 0$

(c): $x + y = 0 = x + (-x)$, so $y = -x$ by (a) with $z = -x$

(d): $(-x) + x = 0$, so by (c) with $-x$ for x: $-(-x) = x$

Proposition 1.15: Cancellation Laws for Multiplication

Proposition 1.15

The axioms for multiplication imply (for $x \neq 0$):

(a) If $xy = xz$ then $y = z$ *(cancellation law)*

(b) If $xy = x$ then $y = 1$ *(uniqueness of 1)*

(c) If $xy = 1$ then $y = 1/x$ *(uniqueness of $1/x$)*

(d) $1/(1/x) = x$ *(double inversion)*

Propositions 1.14 and 1.15: The Parallel Structure

	Addition (1.14)	**Multiplication (1.15)**
Identity	0	1
Inverse	$-x$	$1/x \ \ (x \neq 0)$
(a) Cancel	$x+y = x+z \Rightarrow y=z$	$xy = xz \Rightarrow y=z$
(b) Unique id	$x+y = x \Rightarrow y=0$	$xy = x \Rightarrow y=1$
(c) Unique inv	$x+y = 0 \Rightarrow y=-x$	$xy = 1 \Rightarrow y=1/x$
(d) Double inv	$-(-x) = x$	$1/(1/x) = x$

Proposition 1.16: Further Consequences

Proposition 1.16

The field axioms imply, for any $x, y, z \in F$:

(a) $0x = 0$

(b) If $x \neq 0$ and $y \neq 0$ then $xy \neq 0$

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The field axioms imply, for any $x, y, z \in F$:

(a) $0x = 0$

(b) If $x \neq 0$ and $y \neq 0$ then $xy \neq 0$

(c) $(-x)y = -(xy) = x(-y)$

(d) $(-x)(-y) = xy$

Proof of Proposition 1.16(a): $0x = 0$

Proof of (a)

$$0x + 0x = (0 + 0)x = 0x$$

By 1.14(b): $0x = 0$ $\boxed{0x = 0}$

Key idea: Use the distributive law to turn $0 = 0 + 0$ into a statement about products.

Proof of Proposition 1.16(b): No Zero Divisors

Proof of (b)

Assume $x \neq 0$, $y \neq 0$, but suppose $xy = 0$. Then:

$$1 = \frac{1}{y} \cdot \frac{1}{x} \cdot xy = \frac{1}{y} \cdot \frac{1}{x} \cdot 0 = 0$$

This gives $1 = 0$, contradicting axiom (M4).

Proof of Proposition 1.16(c): $(-x)y = -(xy)$

Proof of (c)

Compute:

$$(-x)y + xy = (-x + x)y = 0y = 0$$

By 1.14(c): $(-x)y = -(xy)$

Similarly: $x(-y) + xy = x(-y + y) = x \cdot 0 = 0$, so $x(-y) = -(xy)$.

Proof of Proposition 1.16(d): $(-x)(-y) = xy$

Proof of (d)

By 1.16(c) and 1.14(d):

$$(-x)(-y) = -\big[x(-y)\big] = -\big[-(xy)\big] = xy$$

$$\boxed{(-x)(-y) = xy}$$

The product of two negated elements equals the product of the originals — not by convention, but by proof.

Ordered Fields

Definition 1.17: Ordered Field

Definition 1.17

An ordered field is a field F that is also an ordered set, such that:

(i) $y < z \implies x + y < x + z$ *(order + addition)*

(ii) $x > 0$ and $y > 0 \implies xy > 0$ *(order + multiplication)*

If $x > 0$: *positive*. If $x < 0$: *negative*.

Example: $\mathbb{Q}$ is an ordered field.

Ordered Field = Field + Compatible Order

Field
axioms (A), (M), (D)

Ordered Set
$<$ is a total order

Ordered Field
(i) order + addition
(ii) order + multiplication

Proposition 1.18: Properties of Ordered Fields

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In every ordered field:

(a) If $x > 0$ then $-x < 0$, and vice versa

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(c) If $x < 0$ and $y < z$ then $xy > xz$

(d) If $x \neq 0$ then $x^2 > 0$. In particular, $1 > 0$

(e) If $0 < x < y$ then $0 < 1/y < 1/x$

Proof of Proposition 1.18(a)

Proof of (a): $x > 0 \implies -x < 0$

If $x > 0$, add $-x$ to both sides of $0 < x$ using 1.17(i):

$$-x = -x + 0 < -x + x = 0$$

so $-x < 0$. If $x < 0$, the same argument gives $-x > 0$.

Proof of Proposition 1.18(b)

Proof of (b): $x > 0,\ y < z \implies xy < xz$

Since $z > y$:

$$z - y > y - y = 0$$

so $z - y > 0$. Since $x > 0$, Definition 1.17(ii) gives $x(z - y) > 0$. Therefore:

$$xz = x(z - y) + xy > 0 + xy = xy$$

$\boxed{xy < xz}$

Proof of Proposition 1.18(c)

Proof of (c): $x < 0,\ y < z \implies xy > xz$

By (a), $-x > 0$. By 1.16(c):

$$-[x(z-y)] = (-x)(z-y) > 0$$

so $x(z-y) < 0$, hence $xz < xy$. $\boxed{xy > xz}$

Proof of Proposition 1.18(d)

Proof of (d): $x \neq 0 \implies x^2 > 0$

Two cases:

If $x > 0$: then $x^2 = x \cdot x > 0$ (by 1.17(ii))

If $x < 0$: then $-x > 0$, so $(-x)^2 > 0$. But $(-x)^2 = x^2$ (by 1.16(d))

Since $1 = 1^2$: $\boxed{1 > 0}$

Proof of Proposition 1.18(e)

Proof of (e): $0 < x < y \implies 0 < 1/y < 1/x$

Step 1: Show $1/y > 0$. If $1/y \leq 0$, then since $y > 0$, part (b) gives $y \cdot (1/y) \leq 0$. But $y \cdot (1/y) = 1 > 0$ by (d). Contradiction.

Step 2: Similarly, $1/x > 0$.

Step 3: Multiply $x < y$ by the positive quantity $\frac{1}{x} \cdot \frac{1}{y}$: $\dfrac{1}{y} < \dfrac{1}{x}$

$\boxed{0 < 1/y < 1/x}$

Summary

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Cancellation laws, sign rules, uniqueness of 0 and 1 all follow from the axioms

An ordered field adds a compatible total order — the setting for inequalities

Squares are positive, $1 > 0$, and reciprocals reverse order

Coming up next: The Real Field — constructing $\mathbb{R}$ with the least upper bound property.